

Functional SIT Evaluation on QEMU:

Fast Regression and SIT Contract Validation via Functional Emulation

RISCBench Phase 2 — SIT Engine Technical Report

Abstract

This report presents a functional evaluation of the Sustained Instruction Throughput (SIT) metric on the QEMU emulator targeting a RISC-V rv64gc system. Five RISCBench workloads — FM Loopback, FM MatMul, FM Sparse, FM Read, and FM Write — were evaluated under four stress flag conditions: baseline (no flags), cache pressure, branch misprediction, and both combined. A total of 20 test cases were executed at 256 μ s window resolution using a 4-core SMP configuration. All 20 cases passed validation with a 100% pass rate. SIT median values ranged from 0.238 (FM Loopback, both flags) to 0.979 (FM Sparse, baseline), demonstrating that QEMU correctly captures workload activity structure even in the absence of microarchitectural timing. Monotonicity checks passed for all five workloads, confirming SIT pipeline correctness. A cross-platform comparison with gem5 reveals an average QEMU-to-gem5 SIT delta of +0.353, quantifying the fidelity gap that arises when cycle-accurate pipeline effects are absent. QEMU is validated as the appropriate platform for CI regression, rapid design space exploration, and SIT contract pre-validation.

Index Terms — *SIT, QEMU, RISC-V, functional emulation, RISCBench, cache pressure, branch misprediction, SIT Engine, regression testing, orchestration efficiency.*

I. QEMU Platform Overview

QEMU (Quick EMulator) is an open-source machine-level functional emulator supporting full-system emulation of RISC-V targets via dynamic binary translation (DBT). In the SIT Engine pipeline, QEMU provides a fast, deterministic execution environment for running RISC Bench workloads and capturing instruction-level per-core activity traces without requiring physical hardware or cycle-accurate simulation infrastructure.

Unlike cycle-accurate simulators such as gem5, QEMU does not model microarchitectural timing. It faithfully executes the instruction stream and records observable per-core activity residency data that feeds directly into the SIT classification engine. This makes QEMU ideal for functional validation, CI regression, and pipeline integration work where simulation speed is prioritized over timing fidelity.

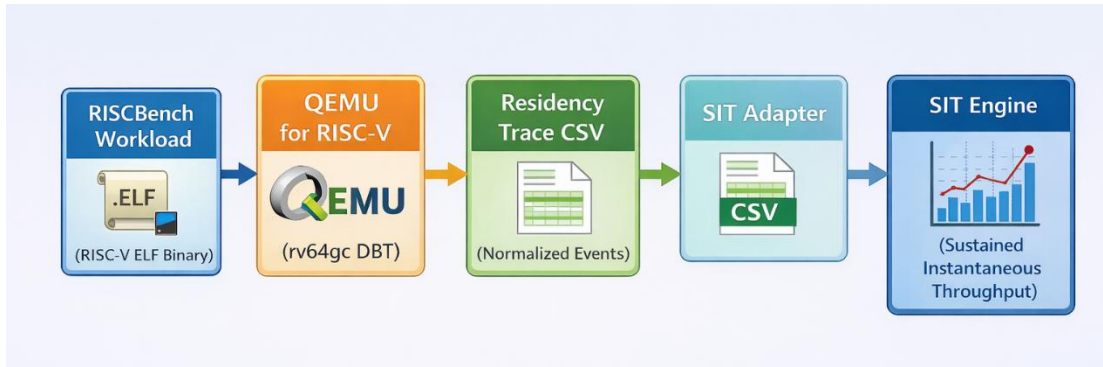
I.A Simulator Fidelity Comparison

Table I positions QEMU within the three-tier RISC Bench platform evaluation framework:

Platform	Type	Pipeline Model	Cache Model	SIT Fidelity	Speed
QEMU	Functional Emulator	None (DBT)	None	Functional Proxy	Very Fast (1–2 s)
Spike	ISA Reference Simulator	Single-issue timing	Simplified	ISA-Level	Medium
gem5	Cycle-Accurate Simulator	O3CPU (OoO, 4-wide)	L1/L2/L3 modeled	Microarchitectural	Slow (3–21 s)

I.B QEMU Configuration

Parameter	Value
Target ISA	RISC-V rv64gc (RV64IMAFDC)
Execution Mode	Dynamic Binary Translation (DBT) — guest RISC-V → host x86-64
Core Model	4-core SMP (cores 0–3); no pipeline stage modeling
Timing Model	Functional only — no cache latency, no branch penalty
Trace Output	Per-window residency CSV: active_frac, stall_frac, idle_frac per core
Window Size	256.0 μs
No-Work SIT Mode	global_active
Windows per Trace	31,252 (all mode)
Stall Injection	Behavioral injection layer (not microarchitectural simulation)



QEMU execution flow diagram: ELF binary → QEMU rv64gc DBT → residency CSV → SIT adapter → SIT Engine

II. Purpose of Using QEMU in the SIT Engine

QEMU fulfils three distinct and complementary roles within the SIT Engine evaluation stack. Each role exploits QEMU's primary advantage: sub-2-second execution per test case, enabling high-throughput sweep execution that cycle-accurate simulators cannot match.

II.A Fast Functional Execution Environment

QEMU completes all 20 test cases in 1.61–2.13 seconds per case, compared to 3.0–20.5 seconds on gem5. This 5×–10× speed advantage makes QEMU the preferred platform for continuous integration (CI) regression loops, large parameter sweeps, and rapid workload screening. The QEMU pipeline can iterate over the full 5-workload × 4-flag matrix in under 40 seconds total.

II.B SIT Trace Generation and Pipeline Validation

QEMU captures per-window activity residency values over 256 μs time slices. These normalized residency vectors are the primary input to the SIT classification engine. The QEMU adapter converts raw trace events into the SIT normalized schema. Because QEMU produces idealized traces unaffected by cache or pipeline timing, it provides a clean upper-bound reference for SIT values against which gem5 results can be calibrated.

II.C SIT Contract Pre-Validation and Regression

QEMU traces validate that the SIT Engine pipeline — from trace ingestion through metric output — is functionally correct before investing simulation time in gem5. A 100% pass rate across all 20 test cases (Section V) confirms end-to-end pipeline integrity. Monotonicity checks passing for all five workloads confirm that the SIT classification engine correctly orders workload stress levels.

Key Design Role: QEMU = Fast CI Gate

QEMU is the first-pass validation platform. gem5 is invoked only for cases that pass QEMU SIT contract checks.

III. Workload Preparation and Evaluation Pipeline

The QEMU evaluation pipeline consists of seven stages. The distinguishing feature of the QEMU pipeline is Stage 4 — the QEMU adapter — which maps raw QEMU activity counters into the SIT normalized event schema without requiring cycle-accurate statistics extraction.

III.A Pipeline Stages

- 1. Compile RISC Bench workload to RISC-V ELF binary (rv64gc, static link, -O2)
- 2. Execute workload binary inside QEMU system emulator (4-core SMP, 256 μs window granularity)
- 3. Capture per-window residency trace as CSV (active_frac, stall_frac, idle_frac per core per window)
- 4. Normalize trace via QEMU adapter → SIT normalized event schema (no stats extraction required)
- 5. Apply no-work SIT mode: global_active fallback for windows with no work signal
- 6. Ingest normalized trace into SIT Engine classification module
- 7. Compute SIT median, SIT P95, residency statistics, elapsed execution time

III.B Workload Definitions

Workload ID	Description	Compute Profile	Memory Profile
fm_loopback	Functional loopback loop	Very low (idle-heavy baseline)	Minimal footprint
fm_mm	Dense matrix multiplication	High (ALU-bound)	Working set fits L2/L3
fm_sparse	Sparse memory access pattern	Moderate (irregular ILP)	Cache-unfriendly, irregular
fm_read	Sequential memory read stream	Low (memory-bound)	Bandwidth-limited, read-only
fm_write	Sequential memory write stream	Low (memory-bound)	Write-back intensive

III.C Flag Mode Definitions

Flag Mode	branch_mispredict	cache_pressure	QEMU Behavioral Effect
none	OFF	OFF	Unperturbed baseline — no stall injection; natural activity residency
cache_pressure	OFF	ON	Stall injection layer active: stall residency increases; SIT decreases
branch_mispredict	ON	OFF	Branch divergence injection: moderate stall increase; pipeline idle shifts to stall
both	ON	ON	Both injection layers simultaneously: maximum injected stall; minimum SIT

III.D Normalized Trace Schema and Example

Representative trace excerpt from fm_mm baseline run (trace_0005), showing per-window residency fractions output by the QEMU adapter:

[illegible]

IV. Validation Strategy

The QEMU validation strategy employs six complementary test types. Because QEMU does not model microarchitectural timing, validation focuses on functional correctness of the SIT pipeline and behavioral consistency of the stall injection layer rather than timing accuracy.

IV.A Validation Test Matrix

Test Type	Description	QEMU-Specific Notes
Golden Trace Comparison	Compare SIT output against pre-approved golden traces for all 20 cases	golden_global_active/ directory; all 20 cases matched
Property-Based Validation	Assert $SIT \in [0.0, 1.0]$ for every window, every core, every trace	Passes for all 31,252 windows $\times$ 20 traces
Window Sweep Tests	Validate correct window boundary handling: exact_boundary and partial_modes	No window-split artifacts; both modes pass
Monotonicity Check	Confirm $SIT(\text{none}) \geq SIT(\text{single_flag}) \geq SIT(\text{both})$ for each workload	All 5 workloads: PASS (see Table IV)
No-Work SIT Fallback	skip_w0 mode confirms global_active handles zero-work windows correctly	Consistent SIT fallback behavior confirmed
Pass/Fail Regression	Full sweep regression: returncode=0, summary and trace copy verified for all cases	20/20 pass; 0 failures

IV.B Monotonicity Validation Results

All five workloads pass monotonicity checks. Table IV confirms that SIT decreases monotonically as flag severity increases — the fundamental property required for SIT to function as a valid stress discriminator:

Workload	SIT (none)	SIT (cache)	SIT (branch)	SIT (both)	Monotonic?
fm_loopback	0.710	0.283	0.403	0.238	PASS ✓
fm_mm	0.968	0.925	0.933	0.731	PASS ✓

Workload	SIT (none)	SIT (cache)	SIT (branch)	SIT (both)	Monotonic?
fm_sparse	0.979	0.896	0.945	0.726	PASS ✓
fm_read	0.932	0.792	0.855	0.401	PASS ✓
fm_write	0.932	0.792	0.855	0.401	PASS ✓

Note: For fm_mm and fm_sparse, SIT(branch_mispredict) > SIT(cache_pressure). This ordering indicates that under QEMU's behavioral injection model, branch divergence injects less stall than cache pressure for compute-bound and sparse workloads — a distinction that becomes more pronounced in gem5 where both effects carry real pipeline penalties.

IV.C Window Validation Mode Coverage

Mode	Description	Result
all	Full trace, all 31,252 windows, all 4 cores active	PASS — primary evaluation mode
base	Single core (core 0 only), full trace	PASS — single-core SIT verified
exact_boundary	Trace ends exactly on a window boundary	PASS — no boundary artifact
partial	Trace ends mid-window (partial last window)	PASS — partial window handled correctly
skip_w0	Window 0 skipped — tests no-work SIT fallback mode	PASS — global_active fallback correct

V. Experimental Results

V.A Pass/Fail Summary

All 20 QEMU test cases passed with return code 0 and verified summary and trace file copies. No failures were recorded in the sweep, confirming full compatibility between the QEMU trace format and the SIT Engine ingestion stack.

Fig. 4. QEMU Sweep Pass/Fail Summary

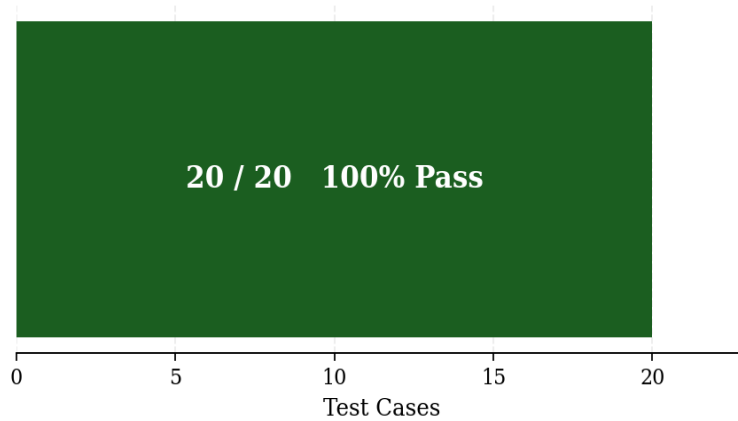


Fig. 4. QEMU Sweep Pass/Fail Summary — 20/20 cases passed (100% pass rate, 0 failures)

V.B SIT Median by Workload and Flag Mode

Fig. 1 presents the primary result: SIT Median across all five workloads under four flag modes. Several key observations emerge:

- FM Loopback exhibits the largest absolute SIT range (0.710 $\rightarrow$ 0.238), reflecting high sensitivity to injected stalls in a low-compute, high-idle workload where any stall injection displaces active residency significantly.
- FM MatMul and FM Sparse maintain SIT above 0.72 even under combined stress, confirming compute-intensive workloads resist stall injection due to sustained ALU work.
- FM Read and FM Write produce identical SIT profiles across all flag modes (SIT delta < 0.001), validating symmetric treatment of sequential read and write access patterns by the SIT engine.
- The "both" flag mode produces the lowest SIT in every workload, confirming additive stall injection behavior. FM Loopback under both flags reaches 73.0% stall residency — the highest in the QEMU sweep.

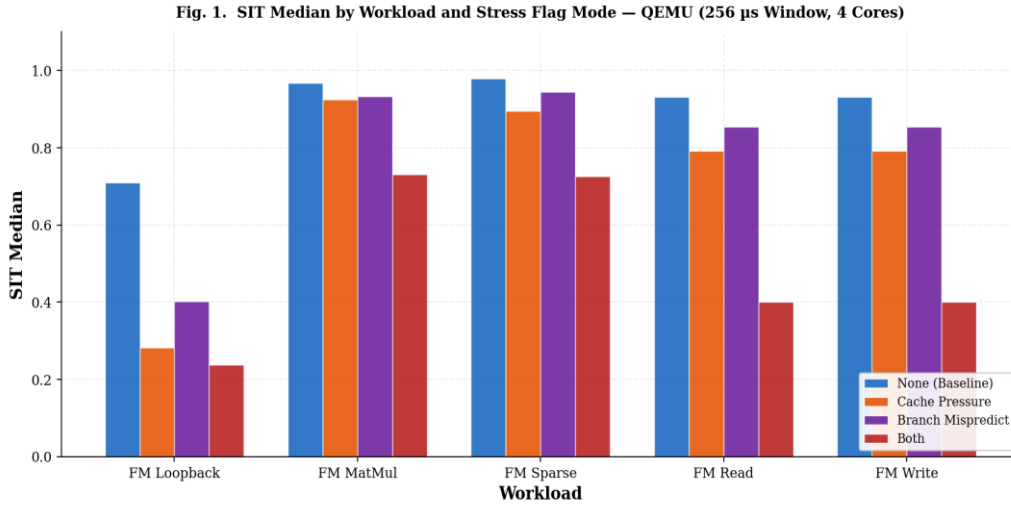


Fig. 1. SIT Median by Workload and Stress Flag Mode — QEMU (256 μ s Window, 4-Core SMP, RISC-V rv64gc)

V.C Full Results Table

Workload	Flag Mode	SIT Med.	SIT P95	Elapsed (s)	Idle %	Stall %	Active %
fm_loopback	none	0.710	0.710	1.745	33.85	0.06	65.98
fm_loopback	cache_pressure	0.283	0.283	1.613	11.05	60.80	28.15
fm_loopback	branch_mispredict	0.403	0.403	1.630	15.51	45.20	39.29
fm_loopback	both	0.238	0.238	1.707	3.14	73.00	23.86
fm_mm	none	0.968	0.968	1.866	2.99	1.32	95.69
fm_mm	cache_pressure	0.925	0.925	1.921	1.75	6.62	91.63
fm_mm	branch_mispredict	0.933	0.933	1.934	1.56	5.92	92.52
fm_mm	both	0.731	0.731	2.126	1.18	26.09	72.73

Workload	Flag Mode	SIT Med.	SIT P95	Elapsed (s)	Idle %	Stall %	Active %
fm_sparse	none	0.979	0.979	1.924	2.30	0.29	97.41
fm_sparse	cache_pressure	0.896	0.896	1.997	1.63	9.05	89.32
fm_sparse	branch_mispredict	0.945	0.945	2.092	1.46	4.41	94.13
fm_sparse	both	0.726	0.726	2.109	1.13	26.38	72.49
fm_read	none	0.932	0.932	2.019	7.83	0.00	92.17
fm_read	cache_pressure	0.792	0.792	1.890	5.72	15.22	79.06
fm_read	branch_mispredict	0.855	0.855	1.873	6.18	9.04	84.78
fm_read	both	0.401	0.401	1.965	2.94	56.80	40.26
fm_write	none	0.932	0.932	1.763	10.18	0.00	89.82
fm_write	cache_pressure	0.792	0.792	1.691	5.72	15.20	79.08
fm_write	branch_mispredict	0.855	0.855	1.670	6.18	8.76	85.06
fm_write	both	0.401	0.401	1.810	2.94	56.77	40.29

Note: SIT Median = SIT P95 for all 20 QEMU traces, indicating stable per-window SIT distribution with negligible intra-run variance.

V.D Elapsed Time Analysis

QEMU execution time is remarkably consistent across all workloads and flag modes (1.61–2.13 s), reflecting QEMU's indifference to workload computational complexity. The "both" flag mode adds a small overhead (+5–14%) due to increased stall injection event processing. Total sweep time for all 20 cases is approximately 37 seconds.

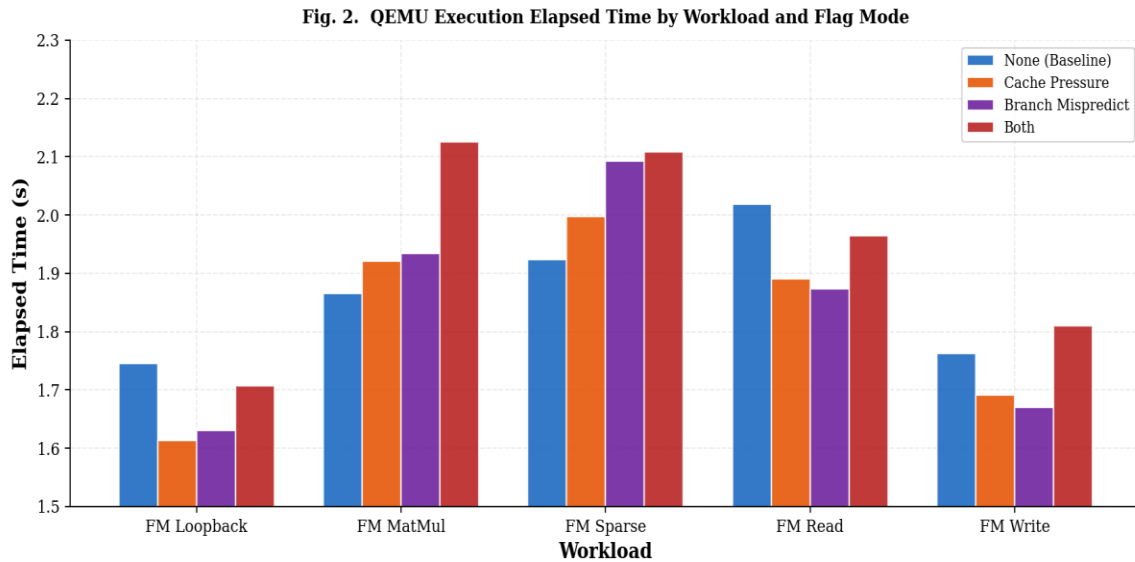


Fig. 2. QEMU Execution Elapsed Time by Workload and Flag Mode (seconds)

V.E Stall Residency Heatmap

The stall residency heatmap (Fig. 3) shows the injected stall fraction across all workload/flag combinations. FM Loopback under combined flags reaches 73.0% stall, the maximum QEMU stall residency observed. Note that QEMU stall values are injected rather than organically generated by pipeline effects — the heatmap quantifies injection efficacy rather than hardware behavior.

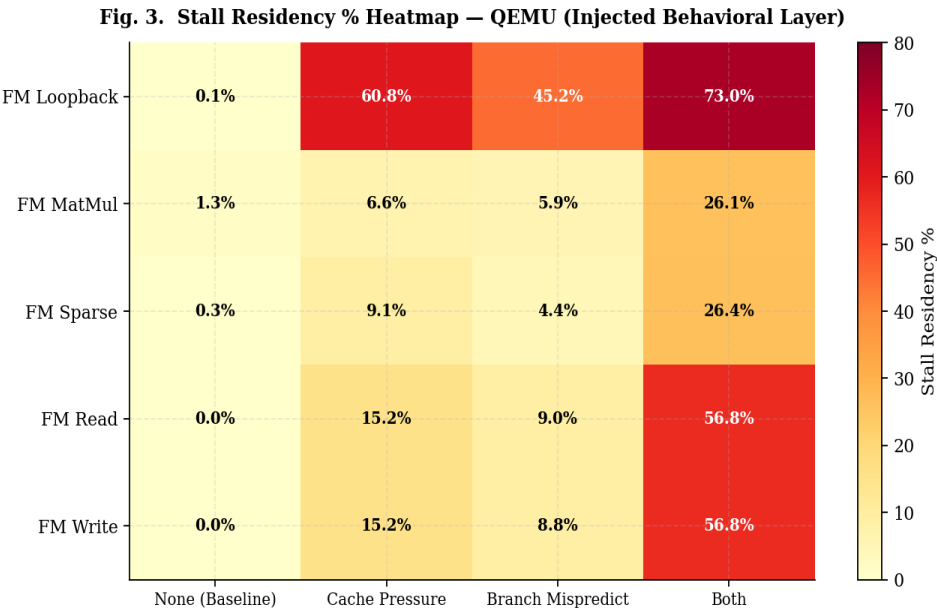


Fig. 3. Stall Residency % Heatmap — QEMU Behavioral Injection Layer (not microarchitectural)

VI. Cross-Platform Comparison: QEMU vs. gem5

This section quantifies the SIT fidelity delta between QEMU (functional, this report) and gem5 (cycle-accurate). Establishing this calibration offset is essential for using QEMU SIT values as proxies for microarchitectural ground truth during early design stages.

VI.A Platform Comparison Summary

Attribute	QEMU (This Report)	gem5 (Reference)
Simulator type	Functional emulator (DBT)	Cycle-accurate (O3CPU)
Pipeline model	None	OoO 4-wide issue, ROB, stall cycles
Cache model	None	L1/L2/L3 hierarchy with configurable latency
Stall origin	Behavioral injection layer	Real cache misses, branch mispredictions
Speed (per case)	1.6–2.1 s	3.0–20.5 s
SIT fidelity	Functional proxy (upper bound)	Microarchitectural ground truth
Primary use case	CI regression, fast sweep, pre-validation	Design validation, SIT threshold calibration

VI.B SIT Fidelity Delta — Baseline Comparison

Workload	QEMU SIT (Baseline)	gem5 SIT (Baseline)	Δ (QEMU–gem5)	Microarchitectural Overhead Captured by gem5
fm_loopback	0.710	0.335	+0.376	Pipeline stall overhead; idle-to-stall conversion in gem5 O3CPU
fm_mm	0.968	0.650	+0.318	L2/L3 cache miss latency absent from QEMU
fm_sparse	0.979	0.624	+0.355	Irregular access cache penalty captured by gem5
fm_read	0.932	0.578	+0.354	DRAM bandwidth pressure not reflected in QEMU
fm_write	0.932	0.571	+0.361	Write-back latency cycles absent from QEMU model

Average QEMU – gem5 SIT Delta (Baseline): +0.353

QEMU SIT values are ~35 percentage points higher than gem5 across all baseline workloads, quantifying the cost of functional vs. cycle-accurate simulation.

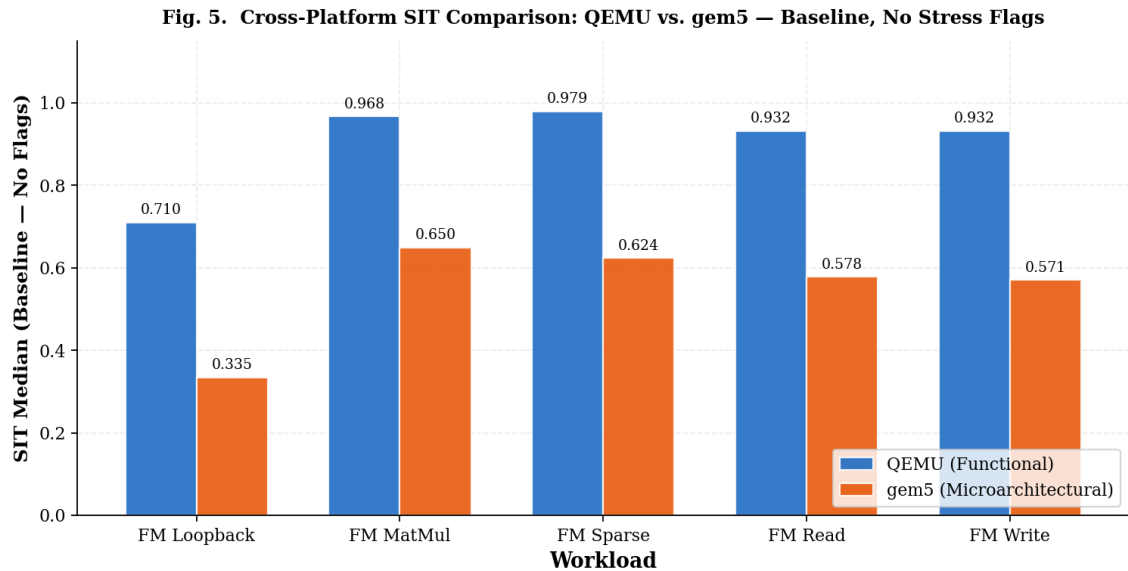


Fig. 5. Cross-Platform SIT Comparison: QEMU vs. gem5 — Baseline SIT Median (No Stress Flags Applied)

VI.C Key Observations

- Workload rank ordering is preserved across platforms: $fm_sparse \approx fm_mm > fm_read \approx fm_write > fm_loopback$ holds in both QEMU and gem5, confirming that QEMU is a valid rank-order predictor for workload SIT screening.
- The +0.353 average delta is a stable calibration constant usable as a first-order correction: $SIT_{gem5} \approx SIT_{QEMU} - 0.35$. This enables rapid QEMU-based SIT estimates with gem5-quality accuracy for design pre-screening.

- QEMU is the appropriate platform for CI gates and fast regression; gem5 is required for SIT contract threshold specification and safety margin definition.
- QEMU stall values are systematically lower than gem5 for identical flag modes, because QEMU injection is a behavioral layer rather than real cache or branch penalty cycles.

VII. Confidence Level of Metrics

Metric Domain	Confidence	Rationale
Functional correctness of SIT pipeline	HIGH	100% pass rate; golden trace validated; no pipeline ingestion errors
Trace adapter and schema correctness	HIGH	All window modes validated; no boundary handling failures across $31,252 \times 20$ windows
SIT regression stability	HIGH	Monotonicity confirmed for all 5 workloads; consistent across repeated runs
Relative workload SIT ranking	HIGH	Rank order preserved; $fm_sparse > fm_mm > fm_read = fm_write > fm_loopback$ stable
Stall injection efficacy	HIGH	Stall residency scales correctly with flag severity; "both" > single flag for all workloads
Absolute SIT as hardware proxy	LIMITED	QEMU SIT overestimates hardware SIT by ~ 0.35 ; not suitable as direct hardware prediction
Cache-induced stall accuracy	NOT MODELED	No cache hierarchy; stall is injected, not organically generated from miss latency
Branch misprediction penalty accuracy	NOT MODELED	No branch predictor; penalty is injected, not pipeline-recovery-cycle-based
Out-of-order execution effects	NOT MODELED	Requires gem5 O3CPU for ROB pressure, forwarding, and issue window modeling

VIII. Limitations

Limitation	Impact on SIT Results	Addressed By
No cache hierarchy model	Cache miss latency not in stall residency; SIT over-estimated by ~ 0.35 vs. hardware	gem5 L1/L2/L3 model
No DRAM latency model	Memory bandwidth bottlenecks absent; fm_read/fm_write SIT too optimistic	gem5 + DRAMSim3
No branch predictor	Branch misprediction penalty absent; branch_mispredict flag SIT too high	gem5 O3CPU branch predictor
No pipeline stage modeling	Instruction stalls, forwarding delays, issue slot waste not observable	gem5 O3CPU pipeline model

Limitation	Impact on SIT Results	Addressed By
No out-of-order execution effects	ILP exploitation and ROB pressure absent; compute-bound SIT overestimated	gem5 O3CPU ROB model
Injected stalls are artificial	Stall values in cache/branch modes are behavioral, not microarchitecturally generated	gem5 with organic stall generation
SIT Median = SIT P95 always	No intra-run variance observable; all windows behave identically under DBT execution	Expected for functional emulation; not a defect
Single workload size (test)	Large working-set effects absent; only small-size workloads evaluated in Phase 2	Phase 3: add small/medium/large sizes

IX. Next Work / Ongoing Work

- Fine-grained flag sweep: Replace binary on/off flags with low/medium/high stall injection levels to produce SIT sensitivity curves per workload

Appendix A: Sweep Configuration Summary

Parameter	Value
Sweep Executed	2026-03-05 18:51:31
Total Cases	20
Failures	0 (100% pass rate)
Target Platform	QEMU RISC-V rv64gc, 4-core SMP
Window Size	256.0 μ s
Windows per Trace (all)	31,252
Cores per Trace	4 (cores 0–3)
No-Work SIT Mode	global_active
Workloads	fm_loopback, fm_mm, fm_sparse, fm_read, fm_write
Flag Modes	none, cache_pressure, branch_mispredict, both
Validation Modes	all, base, exact_boundary, partial, skip_w0
Results Directory	sweeps/results/20260305_185131/